

Original Article

Using Python Programming to Develop Quantity Takeoff Software with a Tkinter Graphical User Interface

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Abstract - This study presents the development and validation of a standalone Quantity Takeoff toolkit for reinforced concrete construction using Python programming and a Tkinter graphical user interface. The application integrates automated modules for reinforcement weight estimation, stirrup quantity calculation with single- and two-zone spacing, concrete volume computation, formwork area estimation, and a bidirectional unit converter, all based on standard engineering formulations such as the $d^2/162$ method. Validation using multiple structural case studies demonstrated zero deviation from conventional manual calculations, confirming the accuracy of the computational framework. A task-based efficiency assessment further revealed that the toolkit reduces calculation time by approximately 70% while significantly minimizing human input errors. The system provides a lightweight, cost-effective, and offline solution suitable for construction engineers, particularly in on-site and small- to medium-scale projects where complex BIM-based tools are not practical, highlighting the effectiveness of compact digital applications in improving productivity and reliability in construction workflows.

Keywords - construction digitalization, formwork estimation, python application, quantity takeoff, reinforced concrete.

1. Introduction

Accurate quantity takeoff is a fundamental process in reinforced concrete construction, forming the basis for cost estimation, material procurement, and construction planning. Errors in estimation may result in financial loss, project delays, and inefficient resource utilization. Traditional methods, including manual calculations and spreadsheet-based approaches, are widely used but remain time-consuming and prone to human error.

Although Building Information Modeling (BIM) offers advanced automation capabilities, its adoption is limited in small-scale projects due to high cost, technical complexity, and computational requirements. Consequently, there is a need for a lightweight and practical digital solution that bridges the gap between manual methods and BIM systems.

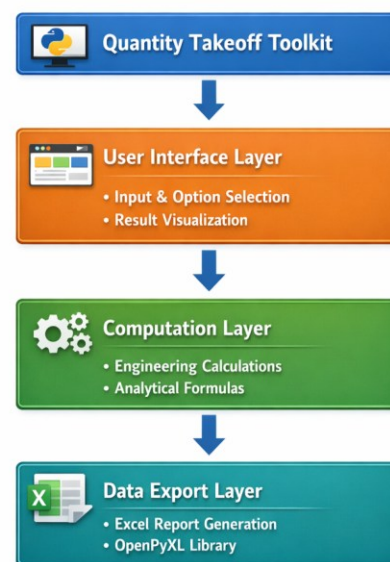
This study proposes a Python-based Quantity Takeoff Toolkit with a Tkinter graphical user interface. The objectives are to develop a standalone application, validate its computational accuracy, and evaluate its efficiency in practical engineering workflows.

2. Materials and Methods

2.1. System Development Framework

The application was developed using Python with

Tkinter for the user interface. The system consists of three layers: User Interface Layer, Computation Layer, Data Export Layer (Excel using OpenPyXL). The tool operates offline and is suitable for field applications.



2.2. Reinforcement Weight Calculation

The unit weight of reinforcement is calculated using:

$$w = \frac{d^2}{162} \quad (1)$$

where:

w = unit weight (kg/m)

d = bar diameter (mm)

The total reinforcement weight is obtained as:

$$L_h = k \times d$$

where:

n = number of bars

L = bar lengths (m)

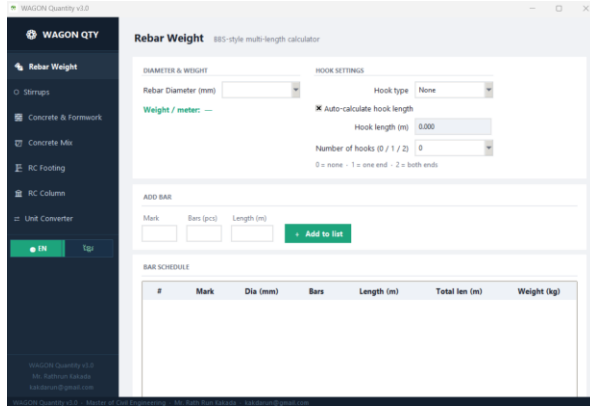
L_h = hook length (m)

nh = number of hooks

Hook lengths are automatically computed as a multiple of the bar diameter:

$$L_h = k \times d$$

Here k depends on hook type (e.g., 10d, 12d, 16d).



2.3. Stirrup Quantity Calculation

2.3.1 Single-Zone Spacing

$$N = \left\lceil \frac{L}{s} \right\rceil + 1$$

Where:

N = number of stirrups

L = member length (m)

s = spacing

2.3.2 Two-Zone Spacing

For structural members with different end and midspan spacing:

$$N = 2 \left(\left\lceil \frac{L_e}{s_1} \right\rceil + 1 \right) + \left(\left\lceil \frac{L_m}{s_2} \right\rceil + 1 \right)$$

Where:

L_e = end zone length

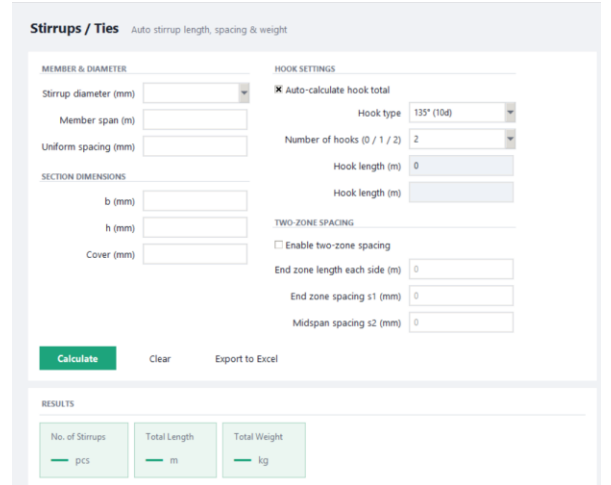
$L_m = L - 2L_e$

s_1 = end zone spacing

s_2 = midspan spacing

The stirrup length is computed from the confined core perimeter:

$$L_s = 2[(b - 2c) + (h - 2c)] + L_{hooks}$$



2.4. Concrete Volume Estimation

Concrete volume is computed using geometric models.

Beam : $V = b \times h \times L$

Slab : $V = L \times W \times t$

Rectangular Column : $V = b \times h \times H$

Circular Column : $V = \frac{\pi D^2}{4} H$

2.5. Formwork Area Estimation

Formwork area is calculated based on exposed surfaces.

$$\text{Beam: } A = (2h \times b)L$$

$$\text{Column: } A = 2(b + h)H$$

$$\text{Wall: } A = 2 \times L \times H$$

2.6. Validation Procedure

To verify the reliability of the toolkit, a series of benchmark problems representing real reinforced concrete elements were prepared. Each case was solved using:

- 1- Manual hand calculation
- 2- Spreadsheet-based calculation
- 3- Developed toolkit

The percentage error was determined as:

$$\text{Error} = \frac{|Q_{\text{tool}} - Q_{\text{manual}}|}{Q_{\text{manual}}} \times 100$$

3. Validation Using Case Studies

To evaluate the accuracy and reliability of the developed Quantity Takeoff Toolkit, a series of structural case studies were conducted. The validation process involved comparing the results generated by the application with conventional manual calculations for different reinforced concrete elements. These elements include beams, columns, slabs, and reinforcement configurations under typical design conditions. The comparison focuses on key output parameters such as reinforcement weight, stirrup quantity, concrete volume, and formwork area. The percentage error was calculated to assess the level of agreement between the two methods.

3.1. Reinforcement Weight Validation

This section presents the validation of reinforcement weight calculations using selected beam and column cases. The parameters considered include bar diameter, number of bars, bar length, and hook conditions. The results obtained from the developed toolkit are compared with manual calculations based on standard engineering formulas. The objective is to verify whether the application can accurately compute reinforcement quantities under varying configurations.

Table 1. Reinforcement Weight Validation

Case	B	N	L	H	M	T	E
Beam	16mm	4	6	2×135°	37	37	0
Column	20mm	8	3.5	none	69	69	0

Note: B = bar diameter; N = number of bar (pcs); L = bar length (m); H = hook (deg.); M = manual(kg); T = toolkit(kg); E = error (%)

3.2. Stirrup Validation

The accuracy of stirrup quantity estimation was evaluated for both uniform (single-zone) and non-uniform (two-zone) spacing conditions. Test cases were selected to represent common structural scenarios in beams. The comparison between manual calculations and toolkit results aims to validate the correctness of spacing logic and quantity determination implemented in the system.

Table 2. Stirrup Validation

Case	Span	Spacing	M	T	E
Beam 1	5.00m	150mm	34	34	0
Beam 2	5.00m	100mm, 200mm	40	40	0

Note: M = manual (pcs); T = toolkit (pcs); E = error (%)

3.3. Concrete Volume Validation

Concrete volume calculations were validated using representative structural elements, including slabs, beams, and columns. The dimensions of each element were defined based on typical design values, and the computed volumes from the toolkit were compared with manual calculations using standard geometric formulas. This validation ensures that the application accurately handles different element types and dimensional inputs.

Table 3. Concrete Volume Validation

Element	Dimension	M	T	E
Slab	4m x 3m x 0.15m	1.8	1.8	0
Beam	0.3m x 0.5m x 6m	0.9	0.9	0
Column	0.4m x 0.4m x 3m	0.48	0.48	0

Note: M = manual (m³); T = toolkit (m³); E = error (%)

3.4. Formwork Area Validation

Formwork area estimation was validated by comparing the calculated surface areas for structural elements such as beams and columns. The toolkit computes formwork based on exposed surfaces, and the results were verified against manual calculations. This comparison confirms the reliability of the implemented geometric logic for practical construction applications. volume calculations were validated using representative structural elements, including slabs, beams, and columns. The dimensions of each element were defined based on typical design values, and the computed volumes from the toolkit were compared with manual calculations using standard geometric formulas. This validation ensures that the application accurately handles different element types and dimensional inputs.

Table 4. Formwork Area Validation

Element	Dimension	M	T	E
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Slab	4m x 3m x 0.15m	14.1	14.1	0
Beam	0.3m x 0.5m x 6m	7.8	7.8	0
Column	0.4m x 0.4m x 3m	4.8	4.8	0

Note: M = manual (m²); T = toolkit (m²); E = error (%)

4. Results and Discussion

4.1. Accuracy of Quantity Takeoff Calculations

The developed toolkit was validated against conventional hand calculations for a series of reinforced concrete elements, including beams, slabs, columns, and reinforcement arrangements. The comparison results (Tables in Section 3) indicate that the computed values from the application are identical to those obtained through manual procedures, resulting in a percentage error of 0.00% for all tested cases.

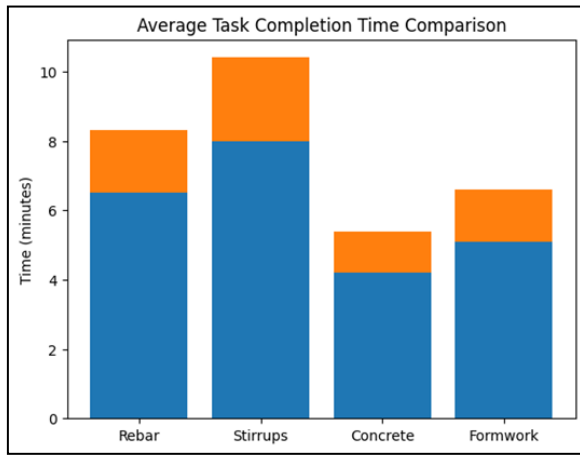


Fig. 1 Average Task Completion Time Comparison

4.2. Time Efficiency Assessment

To evaluate workflow efficiency, a controlled task-based comparison was conducted in which users performed identical quantity takeoff problems using: the traditional manual method, and the developed toolkit.

Table 5. Average Completion Time

Task	M	T	Time Reduction (%)
Rebar calculation	6.5	1.8	72
Stirrup calculation	8.0	2.4	70
Concrete volume	4.2	1.2	71
Formwork area	5.1	1.5	70

Note: M = manual (minute); T = toolkit (minute)

The results show an average time reduction (%) of approximately 70%. This significant improvement is primarily attributed to elimination of repetitive arithmetic operations, automatic unit handling, instant updating of results, removal of intermediate calculation steps. For site engineers who perform multiple operations daily, this reduction translates into substantial productivity gains.

4.3. Error Reduction in User-Based Calculations

In addition to computation time, user-generated errors were recorded during manual and software-assisted workflows. Most manual errors were associated with incorrect spacing conversion (mm → m), missing additional bar length for hooks, miscalculation of stirrup quantity in two-zone regions.

Table 6. Average Completion Time

Method	Average Input Errors per Task
Manual	2.3
Toolkit	0.2

4.4. Functional Integration and Workflow Improvement

A key advantage of the developed system is the integration of multiple functions into a single environment. In conventional practice, engineers typically switch between hand calculations, spreadsheets, unit conversion tools, separate reporting formats. The proposed application consolidates these operations into a unified workflow that performs calculation, displays results immediately and exports structured reports to excel. This integration reduces cognitive load and improves consistency in documentation.

4.5. Suitability for On-Site Engineering Applications

Unlike BIM-based quantity extraction systems, the developed toolkit operates as a lightweight, offline, and low-memory desktop application. This makes it particularly suitable for construction site offices, small and medium contractors, regions with limited access to high-performance computing resources. Its tab-based interface allows engineers to transition quickly between reinforcement, concrete, and formwork calculations without re-entering data into separate tools.

4.6. Comparison with Existing Digital Approaches

While BIM platforms provide automated quantity extraction directly from 3D models, their implementation requires: detailed model preparation, specialized training, high software cost. In contrast, the proposed toolkit requires only geometric input parameters, produces immediate results, can be deployed on standard laptops. Therefore, the system does not aim to replace BIM but to provide an intermediate digital solution between manual calculation and full model-based estimation.

4.7. Limitations of the Developed System

Despite its advantages, several limitations are identified. The application requires manual input of geometric dimensions and does not extract data from drawings. Hook length assumptions are based on predefined multipliers and may vary depending on design codes. The system is not linked to cost databases or scheduling

modules. It is currently limited to common reinforced concrete elements.

4.7. Limitations of the Developed System

The majority of respondents were site engineers and civil engineering students, with most participants having less than one year of professional experience. A significant proportion reported performing quantity takeoff tasks on a daily or monthly basis, confirming the relevance of the survey sample.

- Respondent Role Distribution

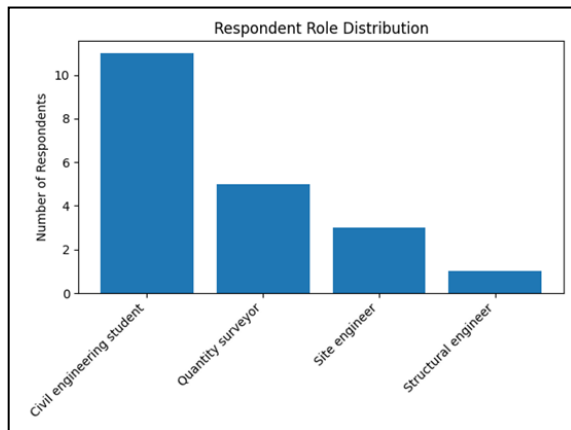


Figure. 2

- Years of Experience Distribution

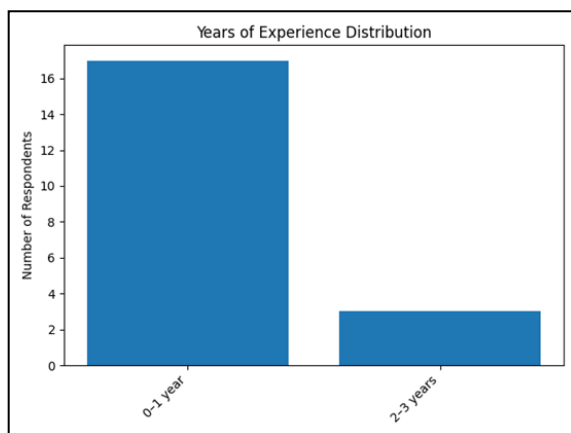


Figure. 3

- Quantity Takeoff Frequency

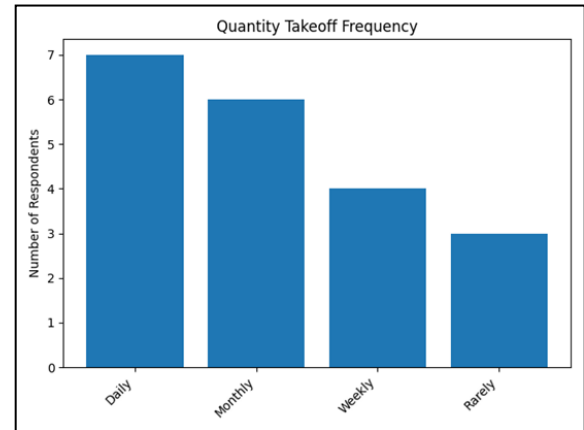


Figure. 4

- Perceived Efficiency and Accuracy

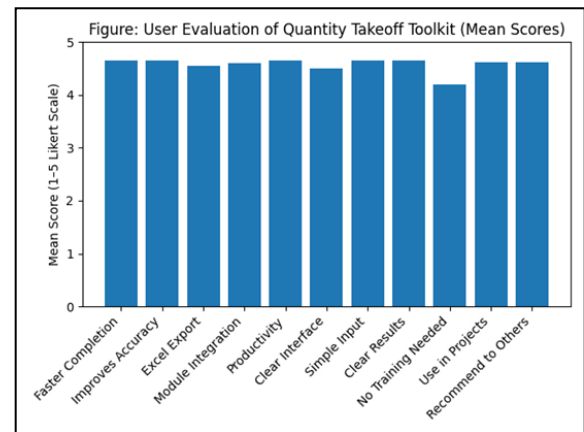


Figure. 5

Overall, the survey results demonstrate strong user acceptance of the developed toolkit. All performance and usability indicators achieved high mean scores (≥ 4.20), with over 94% agreement on productivity improvement and practical applicability. The findings confirm that the system is not only computationally accurate but also user-friendly and suitable for real construction environments.

5. Results and Discussion

This study presented the design and validation of a standalone software Tkinter-based Quantity Takeoff toolkit developed for calculation quantity of reinforced concrete structure. The primary objective was to provide a lightweight, accurate, and practical digital solution that bridges the gap between traditional manual estimation methods and complex BIM-based systems. The developed application integrates multiple calculation modules, including reinforcement weight estimation, stirrup quantity

determination with single- and two-zone spacing, concrete volume computation for common structural elements, formwork area evaluation, and a bidirectional engineering unit converter. In addition, automated Excel report generation was implemented to support standard documentation workflows.

The validation results demonstrated that the outputs produced by the toolkit are fully consistent with conventional hand calculations, with a percentage error of 0.00% for all tested cases. This confirms the correctness of the embedded analytical models and the reliability of the computation engine. More importantly, the time-efficiency assessment revealed that the proposed system reduces calculation time by approximately 70% compared with manual methods. The significant improvement in productivity is attributed to the automation of repetitive arithmetic operations, built-in unit consistency, automatic hook length computation, and structured input control.

The user-based evaluation further showed a substantial reduction in calculation errors. Manual workflows were found to be prone to common mistakes such as incorrect unit conversion, omission of hook lengths, and miscounting of reinforcement numbers. These issues were effectively minimized by the guided interface and automated procedures of the toolkit. As a result, the application not only accelerates the process but also improves the overall reliability and standardization of quantity estimation.

From a practical perspective, the developed system is particularly suitable for on-site engineering applications, small and medium contractors, and construction environments where access to high-end digital infrastructure

is limited. Its offline capability, low computational requirement, and intuitive tab-based interface make it an efficient tool for rapid material estimation, verification of subcontractor quantities, and real-time decision-making during construction. Although the toolkit demonstrates strong performance and practical applicability, several limitations remain. The current system requires manual input of geometric parameters and does not directly extract data from design drawings. Hook length values are based on predefined multipliers and may vary depending on design codes and project specifications. Furthermore, the application is not yet integrated with cost estimation databases, scheduling systems, or BIM platforms. Future development will focus on expanding the functionality of the toolkit by incorporating:

- Automated bar bending schedule generation.
- Drawing-assisted input or model linkage.
- Cost estimation and material optimization modules.
- Mobile and web-based deployment.

In conclusion, the proposed Tkinter-based toolkit provides an accurate, fast, and low-cost digital solution for reinforced concrete quantity estimation. The system significantly improves computational efficiency, reduces human error, and enhances workflow integration without requiring advanced modeling environments. The research demonstrates that compact, purpose-built engineering applications can play a vital role in the digital transformation of construction practices, particularly in resource-constrained and field-based operational contexts.

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Appendix 1 – Software Interface

WAGON Quantity v3.0

WAGON QTY

Rebar Weight

Stirrups

Concrete & Formwork

Concrete Mix

RC Footing

RC Column

Unit Converter

EN

Rebar Weight

BBS-style multi-length calculator

DIAMETER & WEIGHT

Rebar Diameter (mm)

Weight / meter: —

HOOK SETTINGS

Hook type

☒ Auto-calculate hook length

Hook length (m)

Number of hooks (0 / 1 / 2)

0 = none · 1 = one end · 2 = both ends

ADD BAR

Mark

Bars (pcs)

Length (m)

+ Add to list

BAR SCHEDULE

#	Mark	Dia (mm)	Bars	Length (m)	Total len (m)	Weight (kg)
---	------	----------	------	------------	---------------	-------------

Remove selected

Clear list

Export to Excel

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WAGON Quantity v3.0

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EN

₹

Stirrups / Ties

Auto stirrup length, spacing & weight

MEMBER & DIAMETER

Stirrup diameter (mm)

Member span (m)

Uniform spacing (mm)

SECTION DIMENSIONS

b (mm)

h (mm)

Cover (mm)

HOOK SETTINGS

☒ Auto-calculate hook total

Hook type

Number of hooks (0 / 1 / 2)

Hook length (m)

Hook length (m)

TWO-ZONE SPACING

☐ Enable two-zone spacing

End zone length each side (m)

End zone spacing s1

Midspan spacing s2 (mm)

Calculate

Clear

Export to Excel

RESULTS

No. of Stirrups

pcs

Total Length

m

Total Weight

kg

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WAGON Quantity v3.0

WAGON QTY

- Rebar Weight
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Concrete & Formwork

Quantity take-off by element type

ELEMENT INPUTS

Element type: **Footing**

Quantity (pcs): **1**

L (mm):

W (mm):

t (mm):

FORMWORK OPTIONS

☐ Wall: both sides formwork

☐ Beam: 2 sides formwork

☐ Slab: include edge formwork

☐ Footing: include bottom formwork

☒ Column: 3 sides formwork

Concrete: —

Formwork: —

Calculate Clear

BATCH ELEMENT LIST

Element type	Mark	Qty	Dim A (mm)	Dim B (mm)	Dim C (mm)	Option
Beam		1				3 sides (default)

+ Add

#	Mark	Type	Qty	Width (mm)	Length (mm)	Ht / Thk (mm)	Concrete (m³)	Formwork (m²)
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Concrete Mix Quantity

Material quantities per m³ — M18, M20, M25, M30

MIX SELECTION & VOLUME

Concrete Grade: **M30 (1:3:5)** Concrete Volume (m³): **1**

Mix Ratio: 1 : 3 : 5 | Grade: M30 (1:3:5)

Calculate Clear Export to Excel

MATERIAL QUANTITIES

Material	Quantity (kg)
Cement	388.00
Sand	752.00
Gravel	1248.00
Water	200.00
Admixture	1.900

DETAILED BREAKDOWN BY UNIT

Unit	Cement	Sand	Gravel	Water	Admixture
L	269.44	470.00	780.00	200.00	1.900
kg	388.00	752.00	1248.00	200.00	1.900
Bucket (20L)	—	23.50	39.00	10.00	0.095
Bag (50 kg)	7.76	—	—	—	—
m³	0.269	0.470	0.780	0.200	0.0019

Notes: L = Litres · Bucket = 20 L · Bag = 50 kg · m³ = volume of material

QUANTITY PER STRUCTURAL ELEMENT (one-click add)

Element type: **Slab** L (mm): W (mm): t (mm): Qty (pcs): **1** **+ Add element**

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RC Footing Calculator

Rebar · Formwork · Concrete · Lean Concrete

+ Add Footing Type × Remove Current Recalculate All Export All → Excel

F1 Summary

FOOTING PARAMETERS

Footing Mark	F1	Footing Thickness t (mm)	650
No. of Footings (pcs)	1	Lean Concrete Thickness (mm)	100
Footing Length Lx (mm)	1500	Concrete Cover (mm)	75
Footing Width Ly (mm)	2400		

BOTTOM REBAR

Diameter	Spacing X(mm)	Spacing Y(mm)	No.Bars X	No.Bars Y	Take X	Take Y	BarLen X(mm)	BarLen Y(mm)
DB20	150	150	10	16	10	16	1830	2730

Total Bottom: —

MIDDLE REBAR

Diameter	Take No. X	BarLen X(mm)	Take No. Y	BarLen Y(mm)
DB12	4	2850	4	4650

Total Middle: —

TOP REBAR

Diameter	Spacing X(mm)	Spacing Y(mm)	No.Bars X	No.Bars Y	Take X	Take Y	BarLen X(mm)	BarLen Y(mm)
DB12	150	150	10	16	10	16	1638	2538

Total Top: —

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RC Footing Calculator

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+ Add Footing Type × Remove Current Recalculate All Export All → Excel

F1 Summary

TOP REBAR

Diameter	Spacing X(mm)	Spacing Y(mm)	No.Bars X	No.Bars Y	Take X	Take Y	BarLen X(mm)	BarLen Y(mm)
DB12	150	150	10	16	10	16	1638	2538

Total Top: —

RESULTS

Formwork	Concrete	Lean Concrete
m ²	m ³	m ³

REBAR WEIGHT SUMMARY (kg)

RB6mm = —	kg	DB22mm = —	kg
RB8mm = —	kg	DB25mm = —	kg
DB10mm = —	kg	DB28mm = —	kg
DB12mm = —	kg	DB30mm = —	kg
DB14mm = —	kg	DB32mm = —	kg
DB16mm = —	kg	DB36mm = —	kg
DB18mm = —	kg	DB40mm = —	kg
DB20mm = —	kg		

Total Rebar: — kg

Calculate Clear Export to Excel

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WAGON Quantity v3.0

WAGON QTY

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EN

RC Column Calculator

Rebar · Ties · Formwork · Concrete

+ Add Column Type × Remove Current Recalculate All Export All → Excel

C1 Summary

COLUMN PARAMETERS

Column Mark: C1 Width B (mm): 200

Column Type: Rectangular Height H (mm): 200

No. of Columns (pcs): 2 Diameter D (mm):

Column Height (m): 4 ← B × H for rectangular

MAIN BARS

Diameter	No. of Bars	Bar Length (m)*	Total Length (m)	Weight (kg)
DB16	4	—	—	—

* includes lap splice (40d)

TIES / STIRRUPS

Tie Diameter	Zone	Spacing (mm)	No. Ties	Each len (m)	Total len (m)	Weight (kg)
RB8	Beam zone	—	—	—	—	—
	Mid column	—	—	—	—	—
	Slab zone	—	—	—	—	—

Zone length each end (m): 0.8

Spacing @ Beam zone (mm): 100

Spacing @ Mid column (mm): 100

Spacing @ Slab zone (mm): 100

Total Ties: —

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WAGON Quantity v3.0

WAGON QTY

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EN

RC Column Calculator

Rebar · Ties · Formwork · Concrete

+ Add Column Type × Remove Current Recalculate All Export All → Excel

C1 Summary

Spacing @ Beam zone (mm): 100

Spacing @ Mid column (mm): 100

Spacing @ Slab zone (mm): 100

Total Ties: —

RESULTS

Formwork: — m² Concrete: — m³

REBAR WEIGHT SUMMARY (kg)

RB6 = —	kg	DB22 = —	kg
RB8 = —	kg	DB25 = —	kg
DB10 = —	kg	DB28 = —	kg
DB12 = —	kg	DB30 = —	kg
DB14 = —	kg	DB32 = —	kg
DB16 = —	kg	DB36 = —	kg
DB18 = —	kg	DB40 = —	kg
DB20 = —	kg		

Total Rebar: — kg

Calculate Clear Export to Excel

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WAGON Quantity v3.0

WAGON QTY

- Rebar Weight
- Stirrups
- Concrete & Formwork
- Concrete Mix
- RC Footing
- RC Column
- Unit Converter**

WAGON Quantity v3.0
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Unit Converter

Bidirectional, civil-engineering units

CATEGORY: Stress / Pressure

FROM: MPa

TO: psi

⇌ Swap

Type in either box — converts automatically

Clear

WAGON Quantity v3.1

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- Rebar Weight
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- RC Column
- RC Beam**
- RC Slab
- Unit Converter

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RC Beam Calculator

Rebar · Stirrups · Formwork · Concrete

+ Add Beam Type × Remove Current Recalculate All Export All → Excel

B1 Summary

BEAM PARAMETERS

Beam Mark: B1

No. of Beams (pcs): 1

Width B (mm): 200

Height H (mm): 400

Beam Length L (mm): 8000

Concrete Cover (mm): 40

ACI 318 AUTO-LENGTH SETTINGS

Hook Type: 90°(12d)

Lap Splice: 40d

Top bars → L/3 from support (ACI 9.7.3)
Bot bars → 0.7×L mid-span (ACI 9.7.3)
Side bars → full L + lap
Hook extra = multiplier × d at each end

RESULTS

Formwork: m² Concrete: m³

REBAR WEIGHT SUMMARY (KG)

RB6 = — kg RB8 = — kg DB10 = — kg DB12 = — kg
DB14 = — kg DB16 = — kg DB18 = — kg DB20 = — kg
DB22 = — kg DB25 = — kg DB28 = — kg DB30 = — kg
DB32 = — kg DB36 = — kg DB40 = — kg

Total Rebar: — kg

DETAIL BREAKDOWN (ACI AUTO-LENGTH NOTES)

Item	Dia	No.	Bar Len(m)	Total L(m)	Weight(kg)	ACI Note
TOP BARS — SUPPORT (L/3 PER SIDE)						

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WAGON Quantity v3.1 - Master of Civil Engineering - Mr. Rathrun Kakada - kakadarun@gmail.com

RC Slab Calculator

Main Bar · Temp Bar · Concrete

+ Add Slab Type × Remove Current Recalculate All Export All → Excel

S1 Summary

SLAB PARAMETERS

Slab Mark: S1

No. of Slabs (pcs): 1

Slab Area (m²): 20

Slab Thickness t (mm): 150

Concrete Cover (mm): 20

MAIN BAR (SHORT DIRECTION)

Bar Diameter: DB12

Spacing (mm): 150

Span in direction (m): 4

(Short span - bars span this direction)

TEMPERATURE BAR (LONG DIRECTION)

Bar Diameter: DB10

Spacing (mm): 200

RESULTS

Concrete: —

Formwork: —

MAIN BAR RESULT

—

TEMPERATURE BAR RESULT

—

Concrete	Formwork	Main Bar	Temp Bar	Total Rebar
— m ³	— m ²	— kg	— kg	— kg

REBAR WEIGHT SUMMARY (kg)

RB6	RB8	DB10	DB12	DB14	DB16	DB20	DB25
—	—	—	—	—	—	—	—